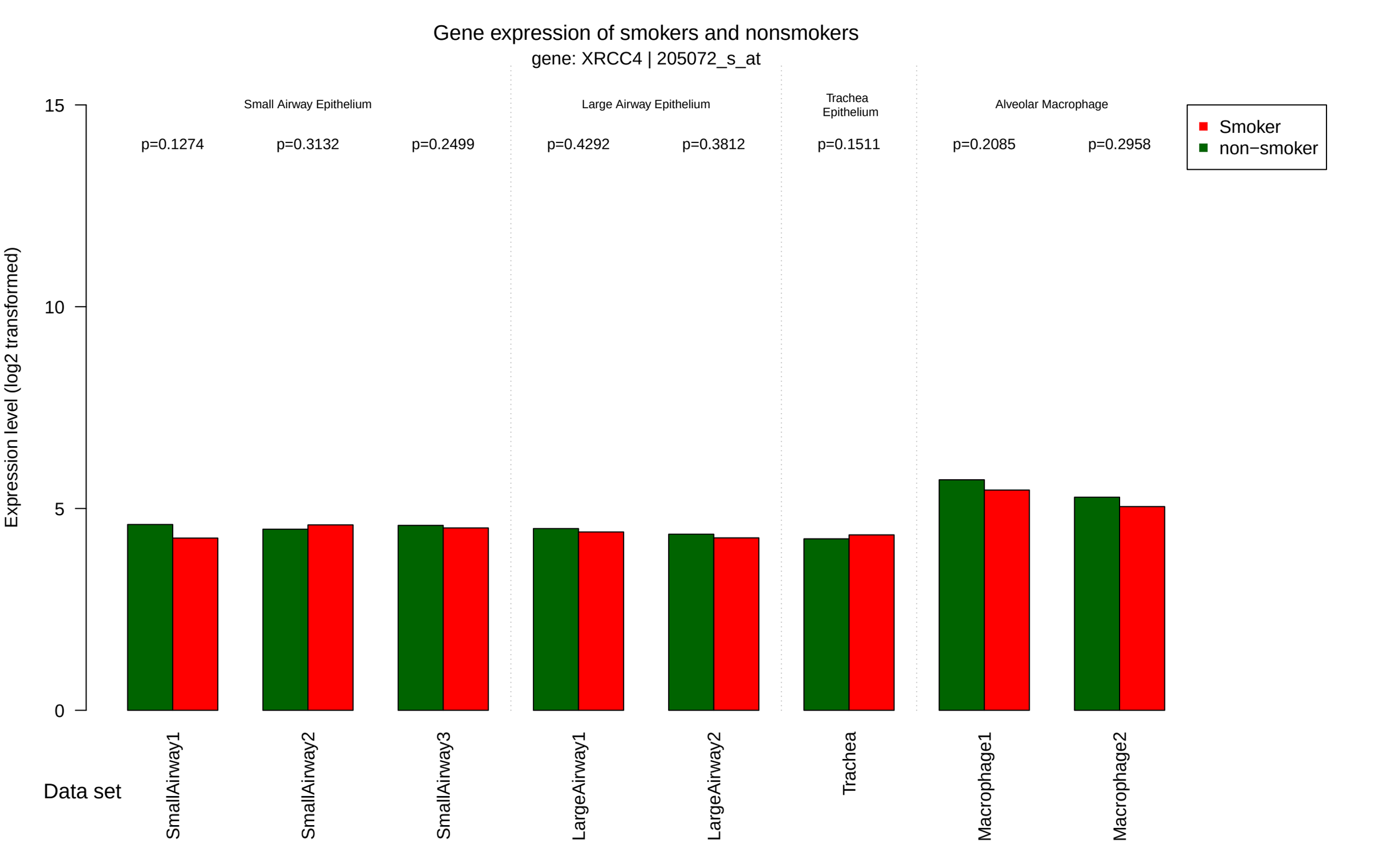
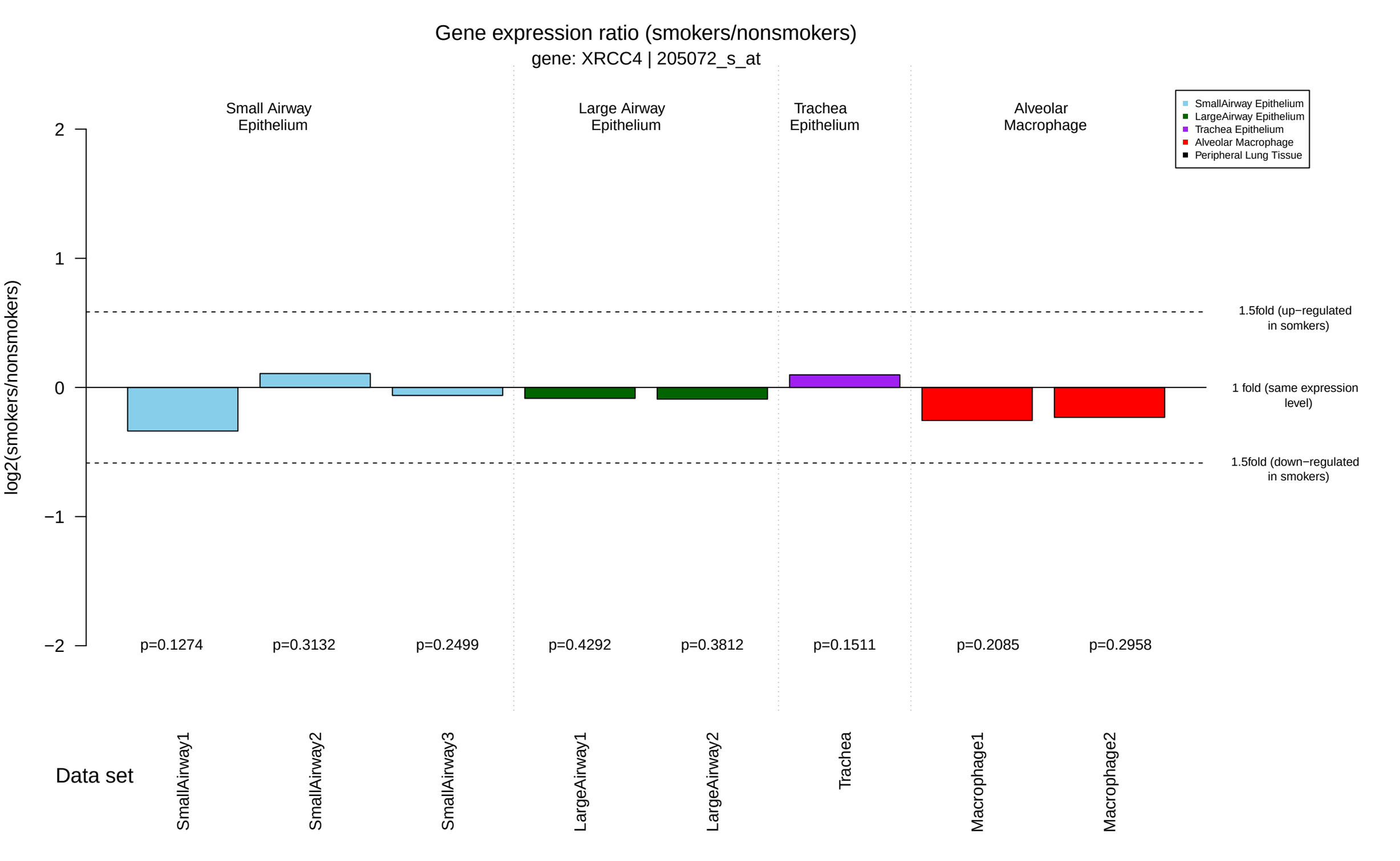
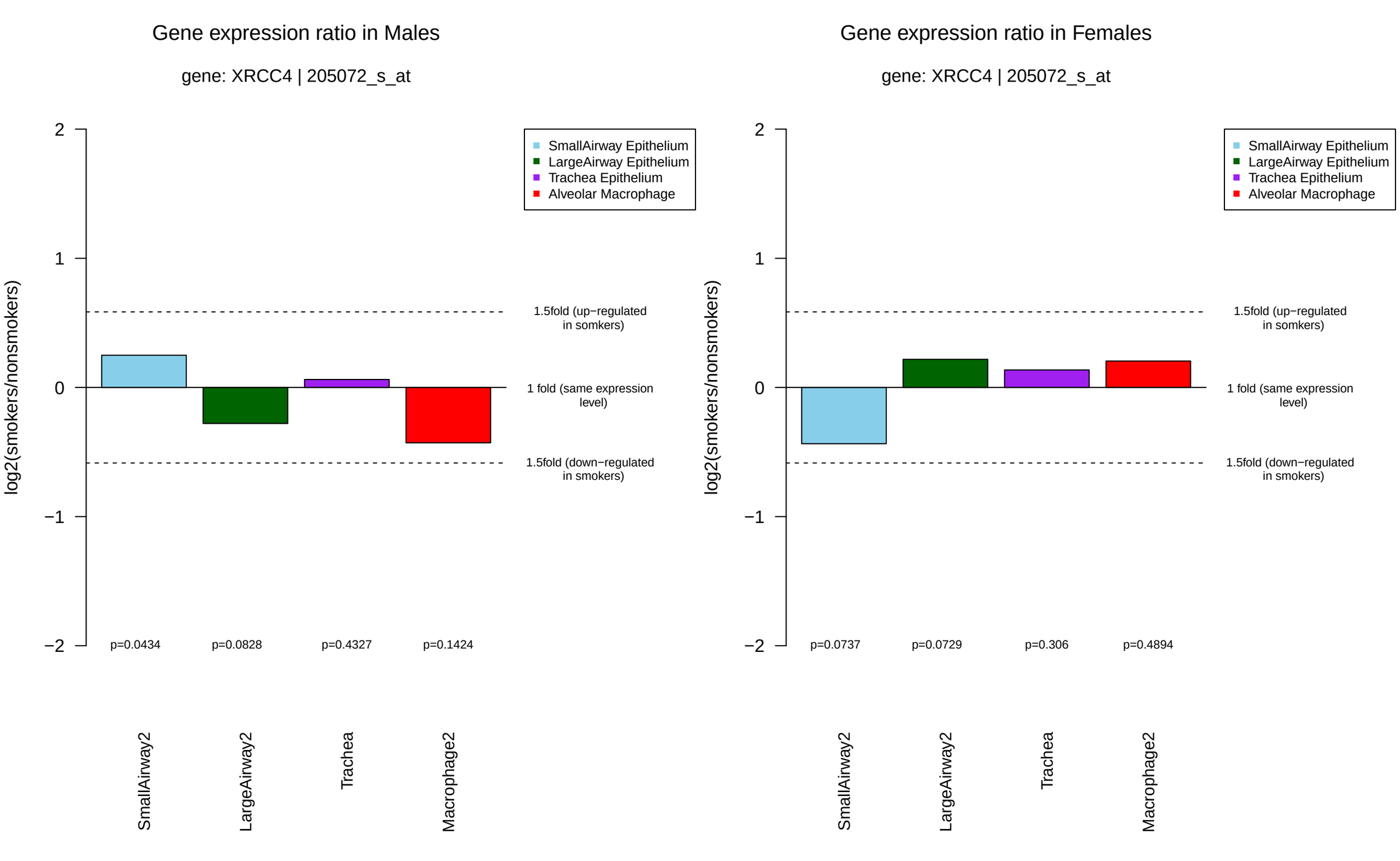
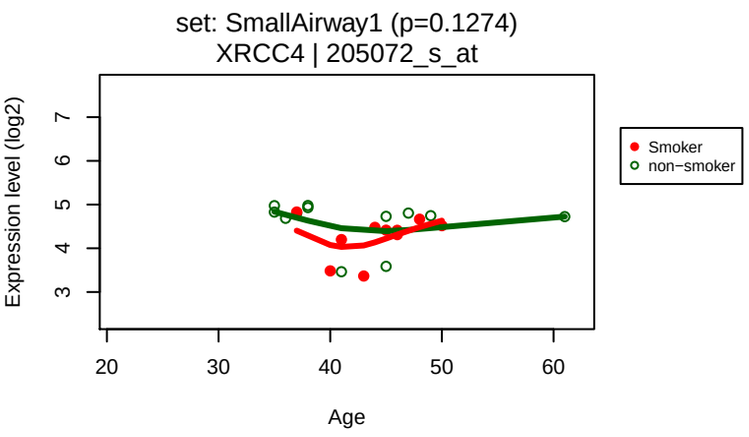
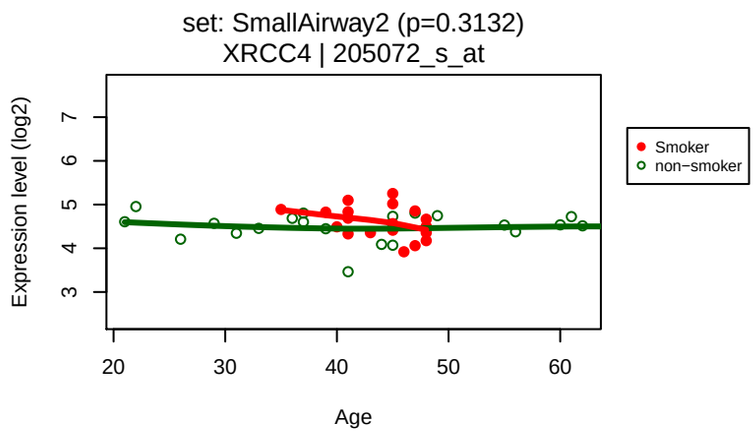
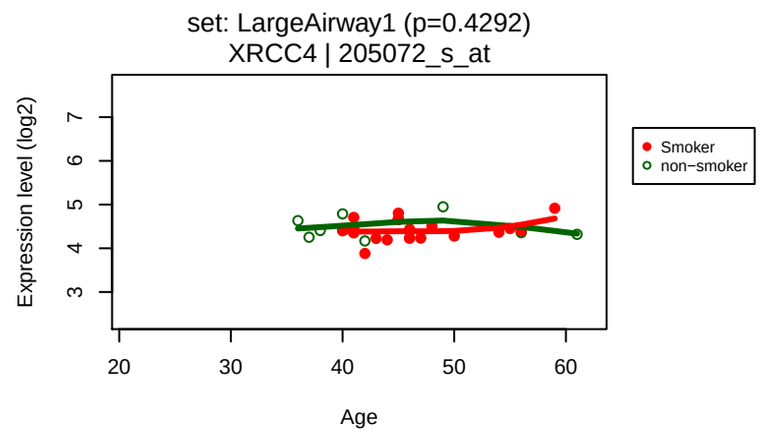

Gene expression of smokers and nonsmokers
gene: XRCC4 | 205072_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1274
p=0.3132
p=0.2499
p=0.4292
p=0.3812
p=0.1511
p=0.2085
p=0.2958
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

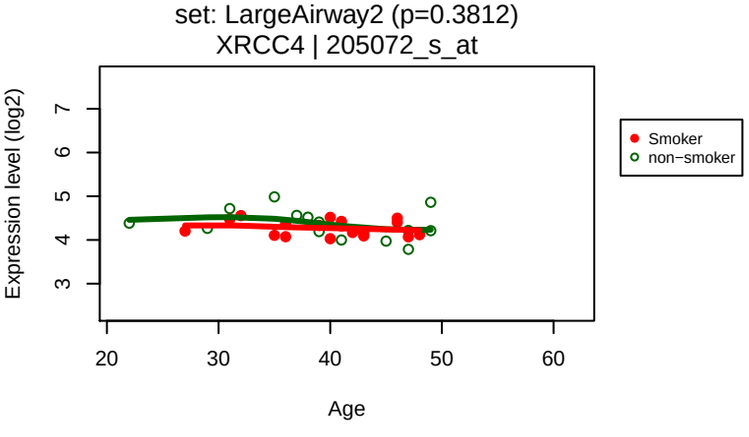

Gene expression ratio (smokers/nonsmokers)
gene: XRCC4 | 205072_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1274
p=0.3132
p=0.2499
p=0.4292
p=0.3812
p=0.1511
p=0.2085
p=0.2958
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

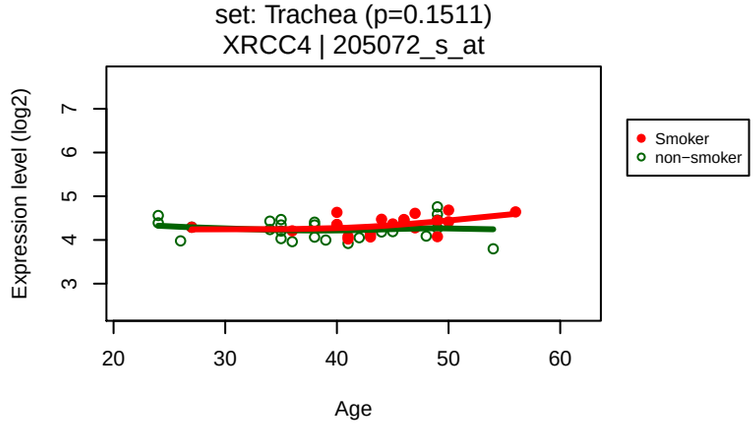

Gene expression ratio in Males
gene: XRCC4 | 205072_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0434
p=0.0828
p=0.4327
p=0.1424
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XRCC4 | 205072_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0737
p=0.0729
p=0.306
p=0.4894
SmallAirway2
LargeAirway2
Trachea
Macrophage2

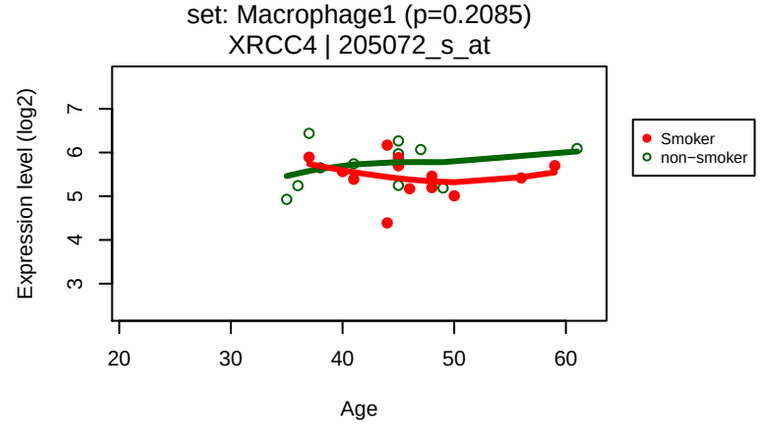

set: SmallAirway1 (p=0.1274)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker

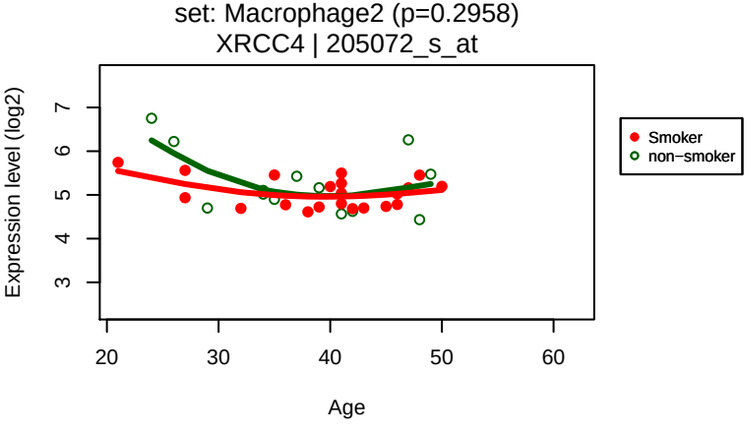

set: SmallAirway2 (p=0.3132)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker

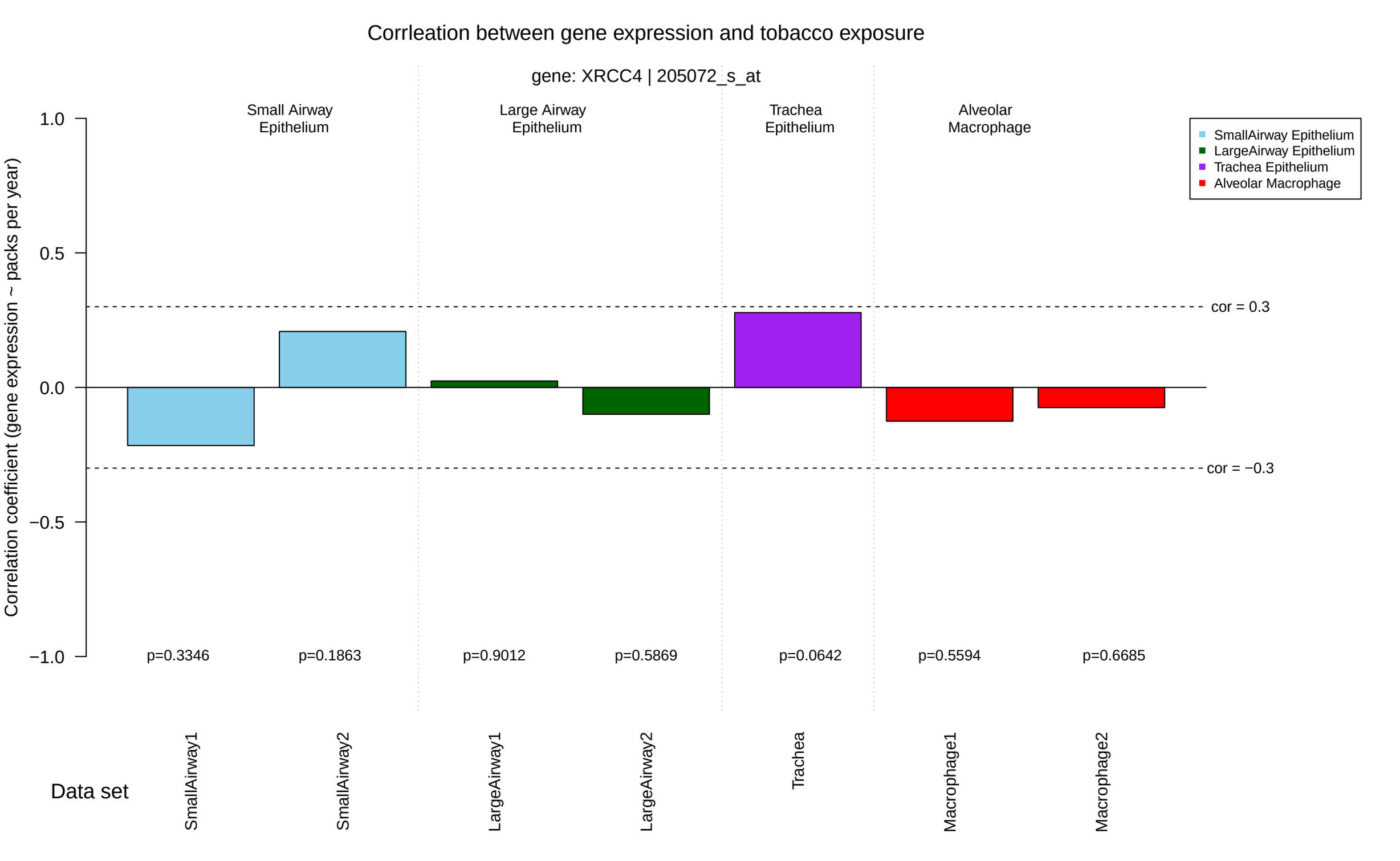

set: LargeAirway1 (p=0.4292)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3812)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1511)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2085)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2958)
XRCC4 | 205072_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XRCC4 | 205072_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3346
p=0.1863
p=0.9012
p=0.5869
p=0.0642
p=0.5594
p=0.6685
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2